3.4.1 Impact on Removing / Adding a portion from / to a given shape

Consider a rectangle of sides 8 *cm* and 12 *cm*.

Length, l = 12 *cm*; Breadth b = 8 *cm*.

Area, A = (l x b) *sq. units.*

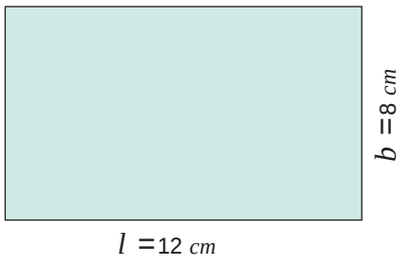
= 12 x 8

= 96 *sq. cm.*

Perimeter, P = 2 (l + b) *units.*

= 2 (12 + 8)

= 40 *cm*.



Find the area and perimeter of the rectangle in the following situations and observe the changes.

Situation 1

A square of side 3 *cm* is cut at a corner of the rectangle.

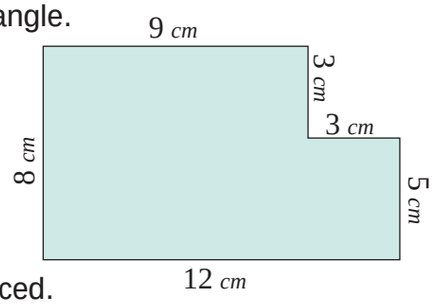
Area, A = (l x b) – (s x s) *sq. units.*

= (12 x 8) – (3 x 3)

= 87 *sq. cm*.

Perimeter, P = (Total boundary) *units.*

= 8+12+5+3+3+9 = 40 *cm*.



The perimeter is not changed. But the area is reduced.

Situation 2

A square of side 3 *cm* is attached to the rectangle.

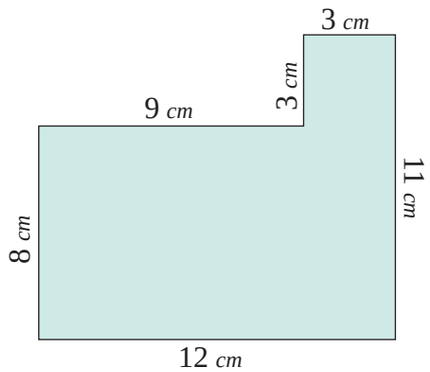
Area, A = (l x b) + (s x s) *sq. units.*

= (12 x 8) + (3 x 3)

= 105 *sq. cm*.

Perimeter, P = (Total boundary) units.

= 8+12+11+3+3+9 = 46 *cm*.



Here both the perimeter and the area are increased.

Example 14 Four identical square floor mats of side 15 *cm* are joined together to form either a rectangular mat or a square mat. Which mat will have a larger area and a longer perimeter?

Solution

Perimeter of a rectangle, P = 2 (l + b) *units.*

= 2 (60+15) *cm*. = 150 *cm*.

Area of a rectangle, A = (l x b) *sq. units.*

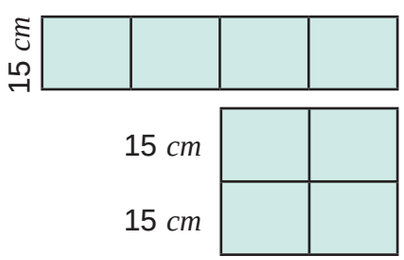
= 60 x 15 = 900 *sq. cm*.

Perimeter of a square, P = (4 × s) *units*

= (4 x 30) *cm* = 120 *cm*

Area of a square, A = (s x s) *sq. units.*

= 30 x 30 = 900 *sq. cm*.



There is no change in their areas. But, the rectangular mat will have longer perimeter.

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